

1- Nitrogen gas has molar mass of 28 kg/kmol. At temperature 1100 K, it has specific enthalpy of 885 kJ/kg and constant pressure specific heat of 1188 J/kg-K.

- (a) Calculate the specific gas constant of nitrogen.
- (b) Find molar specific enthalpy of nitrogen at 1100 K.
- (c) Find the molar, constant pressure specific heat and the specific heat ratio of nitrogen at 1100 K.

2- The chemical formula of nitrogen tetroxide (an oxidizer) is N_2O_4 . Beginning with the masses of the constituent atoms, find its molar mass. Repeat for UDMH (a fuel), which has the chemical formula $(CH_3)_2NNH_2$.

3- Combustion gases have molar mass of 30, and constant pressure specific heat given by the equation $c_p = 0.959 + 1.16(10^{-4})T + 3.65(10^{-8})T^2$ (T in K, c_p in kJ/kg-K). Find and write the equation for c_v in kJ/kg-K. Then, using the equations, calculate the values of c_p and c_v at $T = 1100$ K. Also calculate γ at 1100 K.

4- At a certain location in a flow, the stagnation conditions are 300 kPa and 700 K, the Mach number is 0.8 and the flow area is 0.05 m^2 . $\gamma = 1.3$, $R = 277 \text{ J/kg-K}$.

- (a) Without calculating the static conditions, find the mass flow rate. {Ans.: 21.87 kg/s}
- (b) Calculate the value of A^* from the result of (a). {Ans.: 0.0481 m^2 }
- (c) Now, find the static conditions as well as the velocity at the location. From these values find the mass flow rate.

5- An ideal pump pressurizes hydrogen ($SG = \rho/\rho_{\text{water}} = 0.071$) from 140 kPa to 4 MPa. The flow rate is 45 kg/s. What is the power (sign and magnitude) of the pump? Is the power delivered or absorbed by the pump? {Ans.: $|P| = 2.45 \text{ MW}$ }

6- An ideal turbine is used to drive the pump. Given that the entry and exit stagnation temperatures are 1128 K and 818 K, respectively, calculate the required mass flow rate of the working fluid ($\gamma = 1.3$, $\bar{M} = 10.5$). Also, find the turbine expansion ratio. {Ans.: $\dot{m}_t = 2.3 \text{ kg/s}$ }